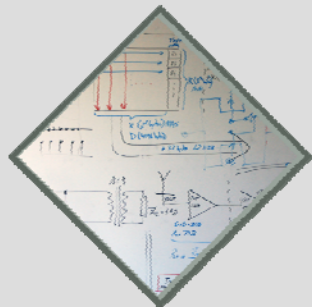
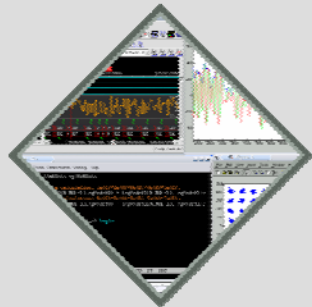
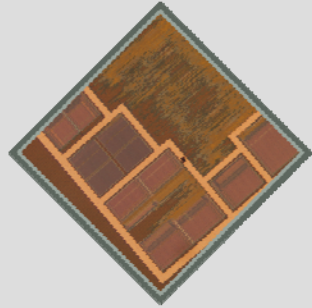
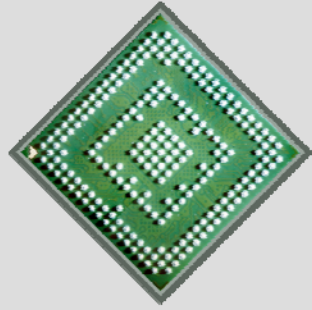




SIDSA

Mobile TV, DVB-H solutions

DVB-H IC Intellectual Property

V1.4.4, July 2006



SIDSA DVB-H OVERVIEW

A DVB-H solution based on Intellectual Property

Intellectual Property products for both the IC and the system industries

- ❑ Solutions for the Semiconductor industry
 - DVB-H and DVB-T Intellectual Property
 - DVB-H Demodulator Intellectual Property and Reference Design
 - DVB-H Software stack

- ❑ Solutions for the System industries: Mobile and Consumer Electronics
 - Receiver module reference design based on:
 - IP Licensee's Silicon
 - SIDSA DVB-H Software

Why Intellectual Property ?

- ❑ Integration is a constant trend in Mobile and Consumer Electronics industries
- ❑ Fewer and fewer companies will have all the building blocks
- ❑ IP is replacing IC as the basic building block as SoC replaces systems
- ❑ SIDSA provides IP and allows customers to concentrate on their existing core knowledge
- ❑ SIDSA concentrates on new demodulator and broadband technology evolution
- ❑ Mobile industry is supplied mainly by semiconductor heavy weights

Why Intellectual Property ? – Revisited –

- ❑ Goal: being perceived as the industry's provider of demodulator technology, first in DVB standards, later in other
- ❑ Supply our customers with state of the art technology, allow them to differentiate themselves with their own added value
- ❑ DVB-H is not the end of the story
 - DMB-T, ISDB-T, FLO... As needed and requested
 - But even more, “4G”: DVB-T2, SSP, ...
- ❑ R&D in OFDM and Single Carrier, Channel estimation, Equalizers....

Intellectual Property is a collaboration Game...

IP Licensees

IP Licensee 1

(US Semiconductor vendor)

IP Licensee 2

(EU Semiconductor vendor)

Reference Design Licensees



Software Partners



lambdastream
SERVICIOS INTERACTIVOS



Tuner Partners



Never stop thinking



MICROTUNE

SHARP

..... be sharp



Technology Partners



TOSHIBA

ARM



DVB-H IP History

- ❑ 2003:
 - DVB-H meetings, IP start, first commercial leads
- ❑ 2004:
 - January 2005 (to mid 2006): Start of WING-TV validation task force project
- ❑ 2005:
 - 3GSM Cannes demo with FreeScale
 - Two IP Licensees
 - August 2005: First customer tape-out
- ❑ 2006:
 - 3GSM in Barcelona with IC and receivers
 - April-July 2006: Two New Tape-Outs
 - June: SDIO and CF design references
 - June: WingTV lab sessions at Nokia, field trials
- ❑ 2007:
 - SIDSA in B21C “Broadcast for the 21st century”. DVB-H, DVB-T2 and DVB-SSP

3GSM Presence

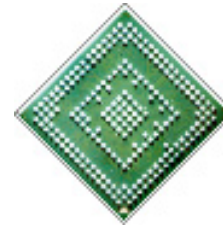
❑ 3GSM 2005 in Cannes

- Demo and Support with FreeScale's ZIF Tuner
- SIDSA's IP implemented in FPGA



❑ 3GSM 2006 in Barcelona

- Two IC Licensees
- IC, SDIO demonstration, DVB-H Broadcast



DVB-H IP Architecture

❑ Dual approach:

- Embedded processor and its software pilot a configurable, highly programmable, demodulator datapath
- Verilog RTL synthesizable in SC gates and plain vanilla memories for best performance, area and power consumption

❑ Pure DSP-based implementations:

- Offer flexibility but ...
- Deliver much worst performance/area/power tradeoffs
- Flexibility is there until the DSP's "SW/HW wall" is reached, when changes become even harder to implement

❑ SIDSA's strategy:

- Begin designs early, participate in standardization and emulate designs
- Be ready with a sound implementation before our competition
- With a Software / Hardware design

Designed from the start for IP use by an IP company

- ❑ No specific libraries needed
 - Simple Standard Cells, RAM, small register banks, (No DPRAM), AFE
- ❑ Memory wrappers around memory blocks for complete technology independence
- ❑ Safe clock management
 - Single clock, safe clock gating in one single top level clock management cell
- ❑ Comfortable clock speed:
 - Typical 62 MHz is simple to achieve, No complex synthesis scripts
- ❑ Industry standards: Cadence, Verilog, Synopsys, ARM ...
- ❑ Flow and code audited by many customers
- ❑ Completed with customers: Equivalency checking, test insertion, formal verification...
- ❑ Simulation verification flow in Verilog & Matlab by verification experts
- ❑ Emulation verification flow by FPGA emulation leaders
- ❑ Record time and smooth tape-out times in all three tape-outs
- ❑ IP is well supported by IC reference Designs and software

DVB-H Partners: RF Tuner support

- ❑ RF tuner companies are Key partners for SIDSA
- ❑ DVB-H direct conversion RF tuners supported
 - FreeScale
 - Infineon
 - Microtune
 - Sharp
 - Broadcom (Athena Semiconductors)
- ❑ Tuners on the pipe:
 - Analog Devices
 - Mirics
- ❑ Modular, software-based RF interface
 - Eases the task of supporting new tuners and supporting multiple tuners
- ❑ SIDSA provides IP with multi-vendor, multiple tuner support.

Consistent performance

- ❑ IP that excels in **all** performance aspects
 - Static and Doppler: MBRAI now, Dibcom performance in October
 - Area: Low area compatible with aggressive industry pricing and RF integration
 - Power: Lowest in the industry with Dibcom
 - Fastest synchronization times in industry
 - TSP: Builds on long tradition of Transport Stream Processor
 - MPE-FEC: Compliant with standards and “de-facto” practices, memory management, designed for extra performance
 - IC design reference platform performance
 - Flexibility where it matters: Demod parameters, RF and power modes
- ❑ No weak points that may compromise system performance and user experience

Static and Doppler Performance

- ❑ True MBRAI compliancy with several tuners
- ❑ Static Performance (AWGN, QEF)
 - IPv1 performance = New theoretical limits + 0.5 dB
 - IPv2 Performance = New theoretical limits - 0.2 dB (*Below* the new theoretical limits!)
- ❑ Doppler Performance (MFER 5, MPEFEC=3/4, 512)
 - IPv1 performance: $F_{dmax} = 110$ Hz (vs MBRAI of 95 Hz) (QPSK, 8K, GI=1/4, CR=1/2)
 - IPv2 performance: $F_{dmax} = 150$ Hz (QPSK, 8K, GI=1/4, CR=1/2)
 - ICI and proprietary algorithms in time and frequency domains
- ❑ AGC and tuner in software enable future growth in tuner management sophistication
- ❑ Strong research in the industry goal: “200 Hz Doppler Wall”
- ❑ SIDSA is an active member of the new MBRAI standardization group



Area

- ❑ IP demodulator Area: 14 sq.mm
 - On “Typical” 0.13 um, 5ML
 - Pessimistic estimation: 20% halo overhead, 140,000 Gates / Sq.mm
 - Demodulator only, from ADC output to Transport Stream Output
- ❑ MPEFEC added area: 1.7 sq.mm with the same technology
 - MPEFEC memory not included
- ❑ Similar to and compatible with typical RF implementations in similar technologies



Power

- ❑ Measured IC Digital Power Consumption:
 - 90 mW “All on” in TSMC .13 um, no low power (High leakage)
 - Includes demodulator and MPEFEC IP + ARM9 and complete bus platform with peripherals streaming four services
- ❑ Demodulator Datapath features built-in fine-grain power control
- ❑ Up to 8 different power modes can be activated by the ARM processor or by HW
- ❑ Processor itself is put to sleep, design can evolve to have multi-supply
- ❑ Vital sync data, pointers and counters are kept for short and perfect sync
- ❑ Shortest sync times in the industry between bursts: 6 to 30 ms vs 150 to 200ms
- ❑ On a typical burst duration, this may represent 50% to 100% *additional* power savings!

Transport Stream and MPE-FEC

- ❑ IP benefits from years of Transport Stream processing at SIDSA
 - Two generations of ARM-based Transport Stream processors on the market
- ❑ Proven MPE-FEC
 - Fought in all the trials, lab tests and shows from 2004 to 2006
- ❑ Most efficient memory management in the industry
 - Removes the memory limitations on consecutive bursts
 - Virtual memory manager
 - Memory is only limited by total bytes sent in one burst
- ❑ TSP and MPE-FEC designed for additional performance
 - Up to 2dB improvement, typical 0,8 dB
- ❑ MPE-FEC and Power Manager designed to work together
 - Automatic or with ARM supervision

DVB-H IP SUPPORT OF STANDARDS

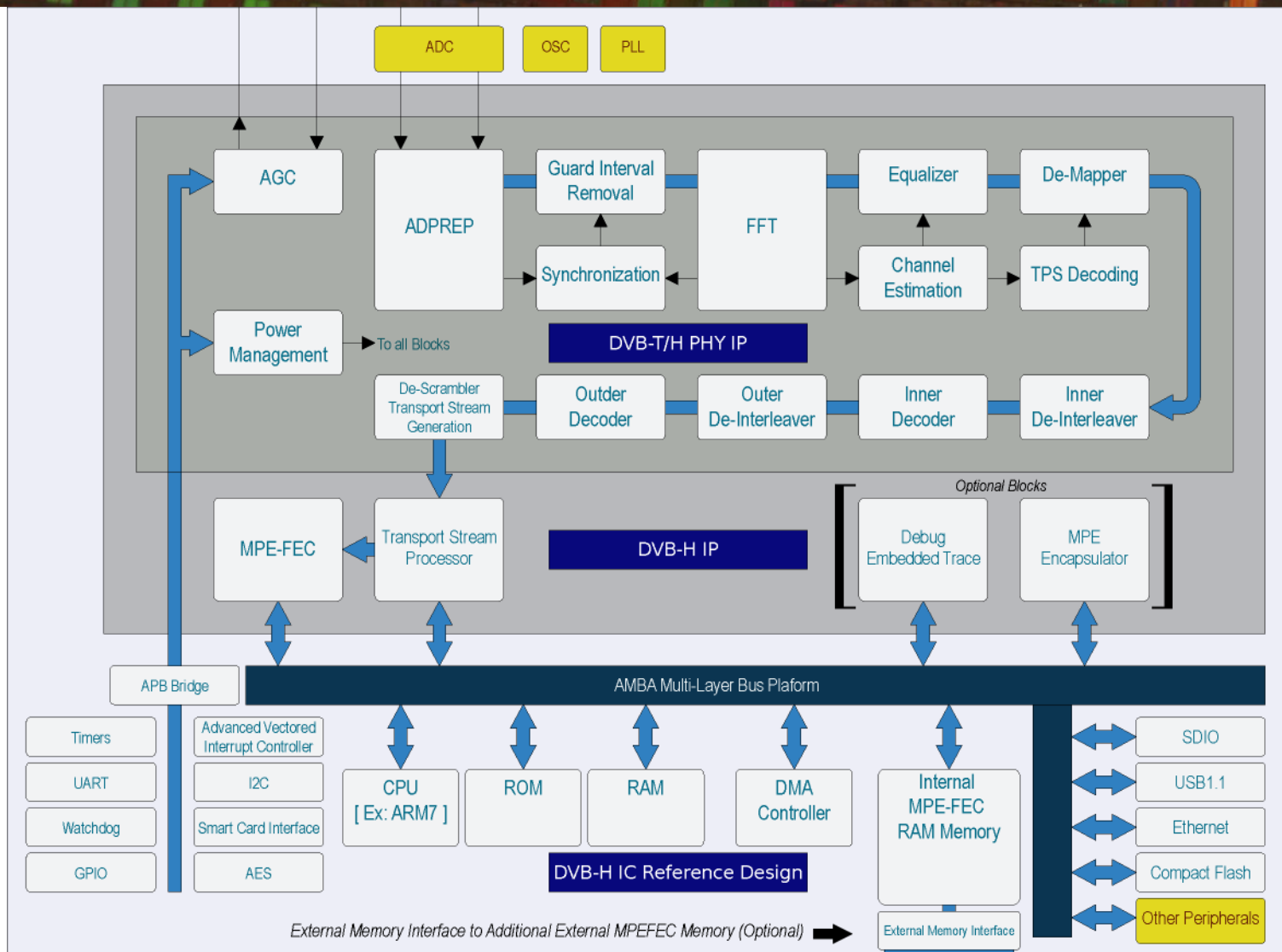
❑ **1: Designed** for Full Support of DVB-H and DVB-T standards

- Hierarchical modulation support
- 2k, 4k and 8k modes
- 8 MHz, 7 MHz, 6 MHz and 5 MHz
- QPSK, 16-QAM and 64-QAM
- Guard interval options: 1/32, 1/16, 1/8 and 1/4
- TPS, MPE-FEC and time slicing
- Advanced power management
- Fast re-synchronization

❑ **2: Verified** in all the verification task force field and lab sessions

❑ **3: Tested** by customers

DVB-T/H Phy IP, DVB-H IP, DVB-H IC Reference Design



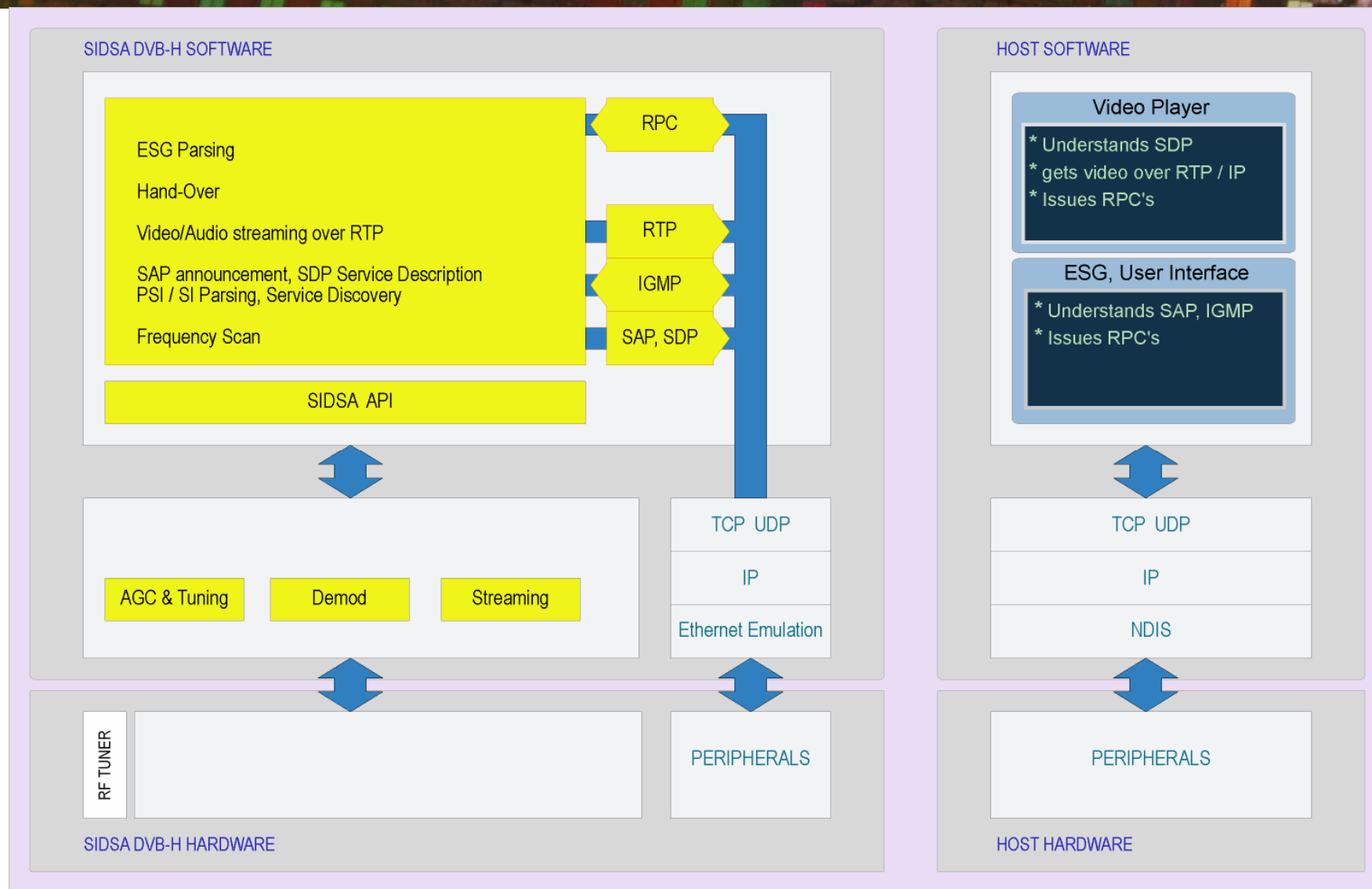
DVB-H Software

- ❑ Software is key to the success of DVB-H, much more than for DVB-T
 - Mobility, hand-over, frequency, IP Datacast ...
- ❑ Low level software included with IP: Drivers and Libraries for:
 - Tuner driver, AGC driver
 - Demodulator
 - Power control
 - Debug
 - MPE-FEC and MTSP (Transport Stream Processor)
- ❑ Optional DVB-H software stack:
 - Hand-over
 - Frequency scan
 - PSI/SI parsing
 - RPC command interface
 - TCP/IP stack and interface
 - Compact Flash and SDIO drivers
- ❑ Special software available: SPP (Service Protection and Purchase)

SIDSA's DVB-H Software Architecture (I)

- ❑ Modular approach based on IP standards
- ❑ Easy integration with Players, OS....
- ❑ Our API is *truly* open: SAP, SDP, IGMP, RTP, RPC...
 - Parse PSI/SI, extract IP address, announce with SAP and SDP
 - Get subscriptions over IGMP
 - Get Video-Audio over RTP IP Packets
 - Control receiver with RPC
- ❑ Example: Only NDIS emulation over SDIO is needed for Windows Mobile 5.0
- ❑ Software running on the ARM may host as well some ESG parsing, IP address management.. Ex: IPV6 to IPV4
- ❑ Interface with other third party middleware is always possible via RPC (Remote Procedure Call)
- ❑ DVB-H Receiver is “IMS READY” !! 

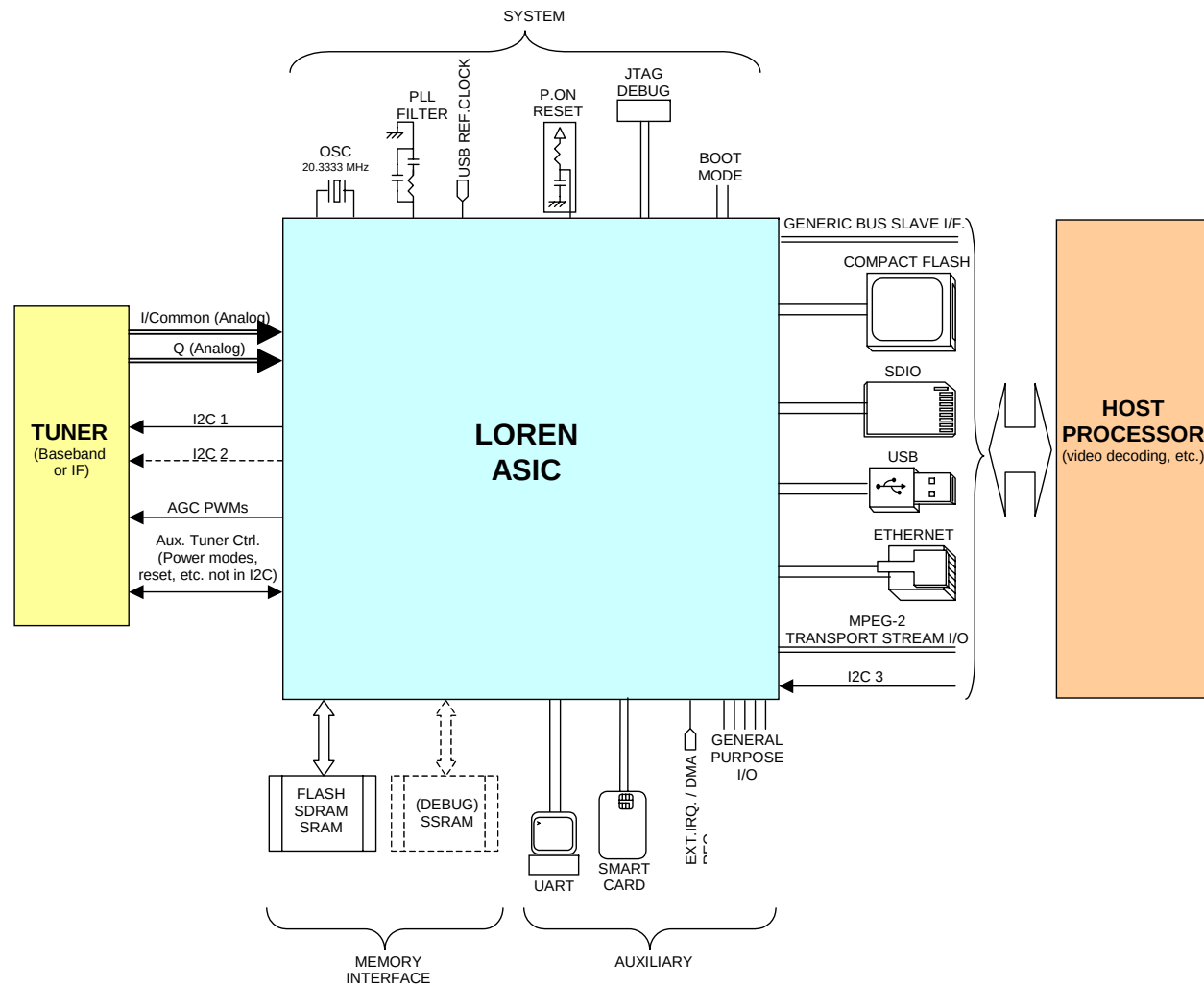
SIDSA's DVB-H Software Architecture (I I)



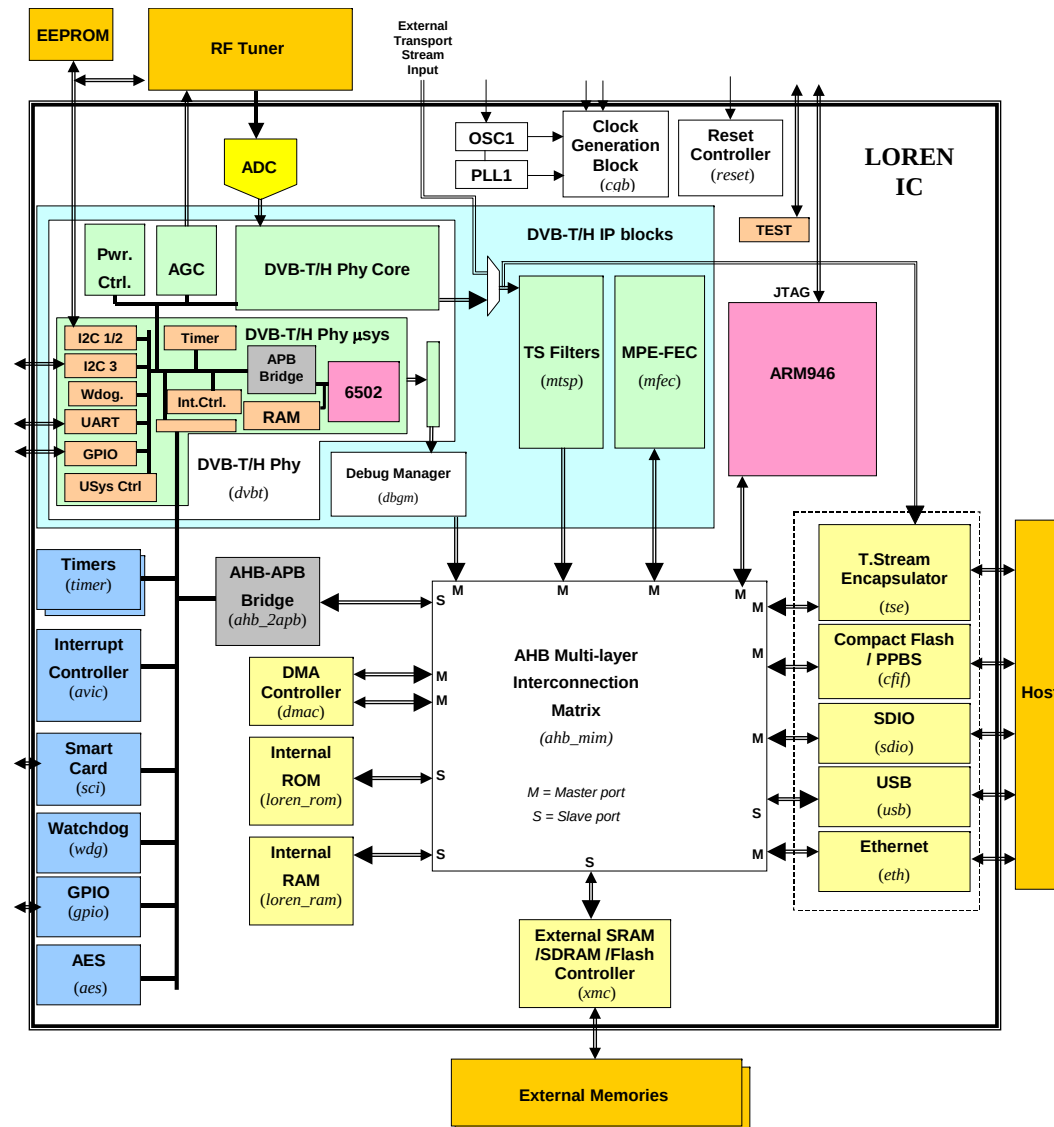
Terminal Reference design

- ❑ DVB-H Receiver designs
- ❑ Based on IC from our licensees and partners
- ❑ Uses our DVB-H software implementation and architecture
- ❑ Specially suited to the consumer industry
- ❑ Deliverables:
 - Schematics, Board Layout, BOM
 - DVB-H software needed
- ❑ IC coming from:
 - SIDSA, in early stage & ramp-up
 - Our customers
- ❑ Licensing: Initial fee and royalties on IC & Software
- ❑ Two licensings done

IC Overview (I)



IC Overview (II)



DVB-H Emulation and Prototyping Systems

Emulation Platform

IP development and evaluation
RF Tuner Integration
Xilinx Virtex 4 100 and 160
Latest IP release



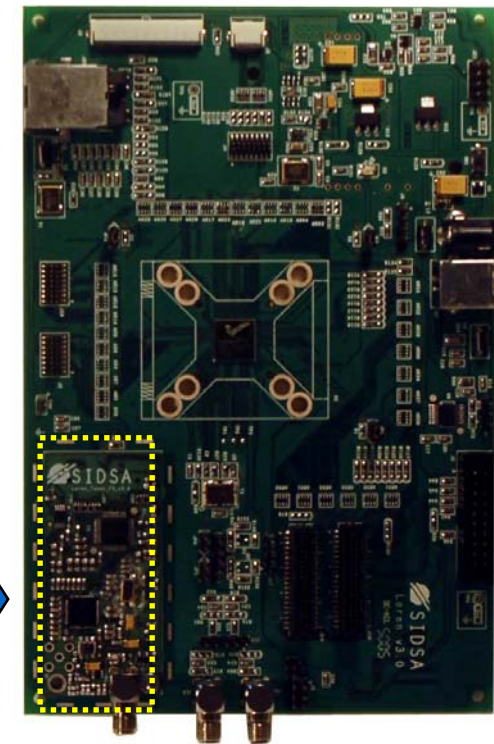
RF Tuner Modules

RF Modules for Different Tuner IC's
Includes ADC
Board BOM and Layout differ
Externally compatible



Professional Receiver

Uses IP-based IC's
USB & Ethernet Outputs



Software debug capabilities

❑ Interactive debug & analysis tool

❑ Monitors:

- Main parameters
- ADC
- AGC
- ADC PreProcessor (ADPREP)
- Synchronization
- FFT
- Channel
- Constellation
- SNR & BER

❑ Uses:

- IC's embedded trace
- Fast DMA mode
- FAST memory access

❑ Standards: PC, Java

The screenshot shows the 'miniLode' software window, titled 'DVB-T/H SIDSA monitoring tool. (c) 2004'. It features a tabbed interface with the following tabs: Configuration, Status, ADC, AGC, AdPrep, Synchronization, FFT, Channel, Constellation, SNR / BER, and IO Settings. The 'Status' tab is currently selected, displaying various monitoring parameters and their current values.

Synchro values

- Fine Time Target :
- Sampling Frequency Offset : ppm
- Frequency Offset : Hz
- Fine Time Offset : us

Synchro detected mode

- FFT mode :
- Guard Interval mode :
- Coarse Time Correlation :

Tps Information

- TPS Error :
- FFT mode :
- Guard Interval mode :
- Modulation mode :
- Code Rate :
- Frame Number :
- Cell ID :

SNR Information

- SNR measured : dB
- Number of samples used :

BER Information

- BER after Viterbi :
- PER after RS :
- Number of packets used :
- Transport Stream Status : VALID

Restart RX Status :

Batch Mode DVB-H Analysis Tool

- ❑ Compatible with SIDSA's Prototyping and emulation systems
- ❑ Batch analysis of performance
- ❑ Performs IC configuration spread analysis
- ❑ Standards: PC, Java, SFQ

